

Calcium

Test Details

METHODOLOGY

Colorimetric

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

Work-up for coma, pancreatitis and other gastrointestinal problems, nephrolithiasis, polydipsia, polyuria, azotemia, multiple endocrine adenomatosis.

Causes of high calcium:

Hyperparathyroidism – look also for high ionized calcium, measured or calculated. Hyperparathyroidism may coexist with other endocrine tumors (multiple endocrine adenomatosis syndromes).

Carcinoma, with or without bone metastases. Humoral hypercalcemia of malignancy (HHM) (tumor-induced hypercalcemia) is seen especially in primary squamous cell carcinoma of lung, head and neck, but other important tumors include primaries in the kidney, liver, bladder, and ovary. It is probably caused by parathormone-like peptides. The most common solid tumors causing bone metastases are primaries in the breast and lung. Other neoplasms may also cause hypercalcemia. Differences between HPT and humoral hypercalcemia of malignancy include low dihydroxyvitamin D, reduced calcium absorption,¹ and the presence of a nonparathyroid tumor. Alkaline phosphatase more than twice its upper limit is more suggestive of **cancer** than of hyperparathyroidism. Especially if there is only a brief duration of symptoms, anemia, hypoalbuminemia, and other findings suggestive of malignant disease, chloride:phosphorus ratio <29 mmol/L, chloride <100 mmol/L, high serum LD (LDH) and/or phosphorus, think first of malignant neoplasm.² **The chloride:phosphorus ratio** is predominantly of value when it is <29 mmol/L, to provide evidence **against** a diagnosis of primary hyperparathyroidism.² Laboratory results which would favor malignancy include anemia, increased LD and alkaline phosphatase, decreased serum albumin and chloride, and chloride:phosphorus ratio <29 mmol/L. **Parathyroid hormone-related protein** was recently purified and identified by molecular cloning as a 141-amino acid peptide with limited homology to PTH itself. Both peptides activate the PTH receptor to produce hypercalcemia. PTH-related protein is now recognized as the cause of hypercalcemia in most solid tumors, particularly squamous, and renal carcinomas.³

Myeloma

Leukemia and lymphoma, especially T-cell⁴ lymphoma/leukemia and Burkitt lymphoma.

Dehydration is an extremely common cause of slight increases of calcium.

Sarcoidosis (a fraction of patients have high serum calcium; usually without low serum phosphorus). More have hypercalciuria.

Chronic hypervitaminosis D. Vitamin A intoxication, isotretinoin (a vitamin A derivative).⁵

Prolonged immobilization (probably uncommon), in patient with increased bone turnover (eg, Paget disease of bone, malignancy, children).

TB, histoplasmosis, coccidioidomycosis, berylliosis

Milk-alkali syndrome: prolonged use of calcium-containing materials and alkali (eg, CaCO_3 or other absorbable alkali ulcer remedies with high milk intake) now rare.

Idiopathic hypercalcemia of infancy (uncommon)

Endocrine: hyperthyroidism, Addison disease, acromegaly, pheochromocytoma (rare cause of hypercalcemia)

Advanced chronic liver disease

Bacteremia

Familial hypocalciuric hypercalcemia⁶ (dominant inheritance); the best test for familial benign hypercalciuria (FBH) is a plot of fasting serum PTH against fasting urine calcium excretion⁷

Aluminum induced renal osteomalacia

Rhabdomyolysis

Several commonly used drugs cause in vivo elevation, including calcium salts, lithium, thiazide/chlorthalidone therapy, other diuretics; vitamins D and A and estrogens (rapid increase in patients with breast carcinoma).

In any case of hypercalcemia, it is desirable to measure magnesium and potassium levels. A helpful mnemonic for the differential diagnosis of the more common causes of hypercalcemia is DCHIMPS (drugs, cancer, hyperparathyroidism, intoxication with vitamin D or A, milk alkali syndrome, Paget disease of bone, sarcoidosis).¹

Causes of low calcium:

Low albumin and low total protein relate to common, usually slight decreases of calcium. The routine method measures **total** calcium, about half of which is bound to plasma proteins. Since the metabolically active form of calcium is the ionized state, the patient's serum protein level should be considered when interpreting a calcium result. For example, a patient's ionized calcium may be normal when the total calcium is elevated in the presence of elevated proteins and, conversely, may also be normal when the total calcium is low and the proteins are low.

High phosphorus: renal insufficiency, hypoparathyroidism, pseudohypoparathyroidism

Vitamin D deficiency, rickets, osteomalacia (Alkaline phosphatase is a test for osteomalacia. Calcium, phosphorus, and alkaline phosphatase can all be normal in osteomalacia.)

Milkman syndrome

Malabsorption or malnutrition with interference with vitamin D and/or calcium absorption

Renal tubular acidosis

Pancreatitis, acute

Dilutional: IV fluids

Bacteremia

Hypomagnesemia

Anticonvulsants and other common drugs, most by in vivo action, can depress calcium. Barbiturates in elderly may cause calcium decrease. Other drugs, including calcitonin, corticosteroids, gastrin, glucagon, glucose, insulin, magnesium salts, methicillin and tetracycline in pregnancy.

Limitations

Sodium citrate, EDTA, and NaF potassium oxalate interfere.

Custom Additional Information

In the differential diagnosis of hypercalcemia serum calcium should be measured on at least three occasions. In **primary hyperparathyroidism** (HPT) parathyroid hormone, serum chloride, and urine calcium are increased. Rarely, in HPT the hypercalcemia is accompanied by a low-normal PTH.⁸ In HPT, calcium rises, then phosphorus falls, then alkaline phosphatase rises. Alkaline phosphatase is usually not more than twice its upper limit in HPT. Measured ionized calcium and calculated ionized calcium may be helpful.

Twenty-four hour urinary calcium is increased in HPT, low in **familial hypocalciuric hypercalcemia** (FHH) which is characterized by hypercalcemia and hypocalciuria. An autosomal dominant, it apparently has no complications. A ratio of renal calcium clearance:creatinine clearance <0.01 suggests this genetic disease. The calcium:creatinine clearance ratio is said to discriminate between FHH and hyperparathyroidism.² Family studies are highly desirable.

Hypocalcemia, then hypercalcemia occur with rhabdomyolysis – induced acute renal failure.^{9,10}

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube; do **not** use oxalate, EDTA, or citrate plasma.

Collection Instructions

Separate serum or plasma from cells within 45 minutes of collection.

Stability Requirements

Temperature	Period
Room temperature	14 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

See table.

Age	Male (mg/dL)	Female (mg/dL)
0 to 10 d	8.6–10.4	8.6–10.4
11 d to 1 y	9.2–11.0	9.2–11.0
2 to 11 y	9.1–10.5	9.1–10.5
12 to 17 y	8.9–10.4	8.9–10.4
18 to 59 y	8.7–10.2	8.7–10.2
>59 y	8.6–10.2	8.7–10.3

Storage Instructions

Room temperature

Patient Preparation

Morning, fasting sample is desirable, since some diurnal variation exists (which may reflect postural changes).

Causes for Rejection

Gross hemolysis; EDTA, oxalate, or citrate anticoagulant; improper labeling

Footnotes

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10. Knochel JP. Serum calcium derangements in rhabdomyolysis. *N Engl J Med*. 1981 Jul 16; 305(3):161-163. PubMed 6894631
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